MANUSCRIPT PREPARATION

6 Steps to Write an Excellent Discussion in Your Manuscript

Table of Contents

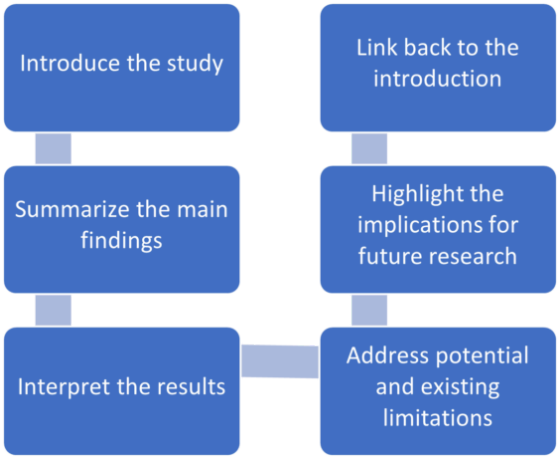
Discussion frame structure
1.Introduction—mention gaps in previous research[1–2]
2. Summarizing key findings—let your data speak[1–2]
3. Interpreting results—compare with other papers[1–2]
4. Addressing limitations—their potential impact on the results[1–2]
5. Implications for future research—how to explore further[1–2]
6. Conclusion

The discussion section in scientific manuscripts might be the last few paragraphs, but its role goes far beyond wrapping up. It's the part of an article where scientists talk about what they found and what it means, where raw data turns into meaningful insights. Therefore, discussion is a vital component of the article.

An excellent discussion is well-organized. We bring to you authors a classic 6-step method for writing discussion sections, with examples to illustrate the functions and specific writing logic of each step. Take a look at how you can impress journal reviewers with a concise and focused discussion section!

Discussion frame structure

Conventionally, a discussion section has three parts: an introductory paragraph, a few intermediate paragraphs, and a conclusion[1]. Please follow the steps below:



1.Introduction—mention gaps in previous research

Here, you orient the reader to your study. In the first paragraph, it is advisable to mention the research gap your paper addresses.

Example: *This study investigated the cognitive effects of a meat-only diet on adults. While earlier studies have explored the impact of a carnivorous diet on physical attributes and agility, they have not explicitly addressed its influence on cognitively intense tasks involving memory and reasoning.*

2. Summarizing key findings—let your data speak

After you have laid out the context for your study, recapitulate some of its key findings. Also, highlight key data and evidence supporting these findings.

Example: *We found that risk-taking behavior among teenagers correlates with their tendency to invest in cryptocurrencies. Risk takers in this study, as measured by the Cambridge Gambling Task, tended to have an inordinately higher proportion of their savings invested as crypto coins.*

3. Interpreting results—compare with other papers

Here, you must analyze and interpret any results concerning the research question or hypothesis. How do the key findings of your study help verify or disprove the hypothesis? What practical relevance does your discovery have?

Example: *Our study suggests that higher daily caffeine intake is not associated with poor performance in major sporting events. Athletes may benefit from the cardiovascular benefits of daily caffeine intake without adversely impacting performance.*

Remember, unlike the results section, the discussion ideally focuses on locating your findings in the larger body of existing research. Hence, compare your results with those of other peer-reviewed papers.

Example: *Although Miller et al. (2020) found evidence of such political bias in a multicultural population, our findings suggest that the bias is weak or virtually non-existent among politically active citizens.*

4. Addressing limitations—their potential impact on the results

Discuss the potential impact of limitations on the results. Most studies have limitations, and it is crucial to acknowledge them in the intermediary paragraphs of the discussion section. Limitations may include low sample

size, suspected interference or noise in data, low effect size, etc.

Example: *This study explored a comprehensive list of adverse effects associated with the novel drug 'X'. However, long-term studies may be needed to confirm its safety, especially regarding major cardiac events.*

5. Implications for future research—how to explore further

Locate areas of your research where more investigation is needed. Concluding paragraphs of the discussion can explain what research will likely confirm your results or identify knowledge gaps your study left unaddressed.

Example: *Our study demonstrates that roads paved with the plastic-infused compound 'Y' are more resilient than asphalt. Future studies may explore economically feasible ways of producing compound Y in bulk.*

6. Conclusion—summarize content

A good way to wind up the discussion section is by revisiting the research question mentioned in your introduction. Sign off by expressing the main findings of your study.

Example: *Recent observations suggest that the fish 'Z' is moving upriver in many parts of the Amazon basin. Our findings provide conclusive evidence that this phenomenon is associated with rising sea levels and climate change, not due to elevated numbers of invasive predators.*

A rigorous and concise discussion section is one of the keys to achieving an excellent paper. It serves as a critical platform for researchers to interpret and connect their findings with the broader scientific context. By detailing the results, carefully comparing them with existing research, and explaining the limitations of this study, you can effectively help reviewers and readers understand the entire research article more comprehensively and deeply[1–2], thereby helping your manuscript to be successfully published and gain wider dissemination.

In addition to keeping this writing guide, you can also use Elsevier Language Services to improve the quality of your paper more deeply and comprehensively. We have a professional editing team covering multiple disciplines. With our profound disciplinary background and rich polishing experience, we can significantly optimize all paper modules including the discussion, effectively improve the fluency and rigor of your articles, and make your scientific research results consistent, with its value reflected more clearly. We are always committed to ensuring the quality of papers according to the standards of top journals, improving the publishing efficiency of scientific researchers, and helping you on the road to academic success.